How to collect Training Data?

Collecting training data is no doubt a tedious task but here are a few ways to make your work easier:

- **Scraping from the web**

Manually downloading images takes a long time due to the amount of human work involved. A good substitute is to program a web scraper or use a web scraping service to download images and text. Annotation still has to be done manually or crowdsourced. LabelMe is a GUI which allows you to manually label images. Amazon Mechanical Turk is an excellent crowdsourcing platform

- **Third-party organizations**

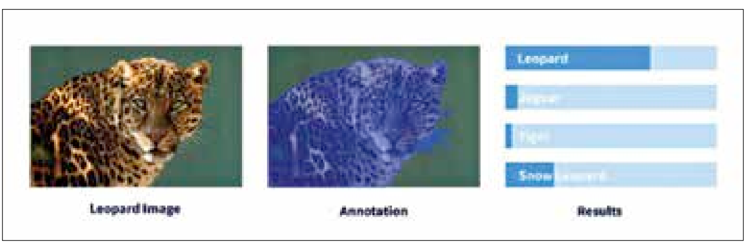
Start-ups are offering data as it has become a prized commodity in the machine learning business. Given the nature and objective of the dataset, the third parties provide a well-designed training dataset.

- **Pre-trained networks**

Publicly available datasets are the cheapest source of bulk data. The datasets used for benchmarking are certainly reliable and huge (Open Images dataset with over 15 million images labelled with bounding boxes from 600 categories). There are open-source models trained on these datasets. Using transfer learning, we can fine tune the pre-trained network to suit our own needs. The accuracy of these models also exceed traditional models.

- **The Snowball effect**

We can use an average performing model as a baseline to collect more data. The data, in turn, is used to train the model. This model will perform better on a task than the general pre-trained model it is fine-tuned for a



Labelling improves performance.